

NYUTABARU AB, Japan

WMO: 478540

Lat: 32.084N

Lon: 131.451E

Elev: 259

StdP: 14.56

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.0	32.2	14.4	11.5	38.9	17.4	13.4	39.5	26.6	47.5	24.2	48.2	7.2	270	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.1	91.2	78.3	88.2	78.6	86.4	78.3	81.2	85.7	80.2	85.2	79.2	84.2	11.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.5	160.1	83.6	79.0	151.9	82.9	78.4	148.6	82.6	45.2	85.4	44.1	86.7	43.0	84.5	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.8	18.2	15.2	24.7	95.6	2.0	2.3	23.3	97.3	22.1	98.6	21.0	99.9	19.6	101.5		
(5)			22.0	83.6	1.9	1.8	20.6	84.9	19.5	86.0	18.4	87.0	17.1	88.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	45.0	47.3	52.5	60.4	67.6	72.7	80.2	81.1	76.3	68.0	58.4	48.9
	DBStd	13.33	5.41	6.29	5.76	5.13	3.60	4.09	3.34	2.61	3.88	4.84	6.01	6.00
	HDD50	430	172	118	39	2	0	0	0	0	0	0	5	94
	HDD65	2403	619	497	389	151	15	0	0	0	0	25	208	499
	CDD50	5279	18	42	116	314	544	681	935	964	790	557	258	60
	CDD65	1778	0	0	1	13	95	232	470	499	340	117	11	0
	CDH74	13580	1	1	3	41	263	1077	4437	5007	2349	386	15	0
	CDH80	3779	0	0	0	5	28	201	1440	1597	477	30	1	0
Wind	WSAvg	6.9	7.7	7.3	7.5	7.2	6.5	6.4	7.0	7.0	6.5	6.4	6.3	7.2
Precipitation	PrecAvg	97.2	2.7	3.8	6.1	7.6	8.9	19.4	14.2	10.4	12.0	6.3	4.0	2.6
	PrecMax	155.6	6.8	7.1	16.1	13.5	19.6	33.9	38.1	21.6	23.8	18.4	10.1	6.2
	PrecMin	61.3	0.0	1.5	1.9	2.2	3.0	6.5	2.9	1.7	2.4	0.5	1.2	0.1
	PrecStd	18.2	1.5	1.8	2.8	2.7	3.5	6.9	8.1	5.2	6.0	4.5	2.2	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.6	69.6	72.0	78.8	83.9	89.7	94.7	94.9	89.9	82.7	75.5	68.2
		MCWB	56.8	61.2	60.6	63.7	69.6	77.2	77.8	79.2	78.2	73.6	66.8	59.8
	2%	DB	60.6	64.6	68.2	73.7	79.0	85.7	91.3	91.2	86.3	80.4	73.0	64.5
		MCWB	53.7	58.1	58.8	63.7	67.7	77.3	78.1	78.6	77.7	72.2	64.9	57.2
	5%	DB	57.4	60.9	65.2	71.6	76.9	82.1	89.2	88.2	84.4	77.4	70.1	61.2
		MCWB	50.3	54.7	56.8	62.4	67.7	75.4	78.6	79.0	76.7	69.7	63.1	54.9
	10%	DB	55.2	57.6	62.6	69.6	74.9	79.1	86.4	86.4	82.6	75.6	68.1	59.1
		MCWB	48.8	51.3	55.5	61.9	66.4	73.2	78.4	78.7	75.1	68.2	61.7	52.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.1	63.6	64.7	70.5	75.0	79.8	82.6	83.2	82.8	77.8	70.5	64.0
		MCDB	63.1	67.3	69.7	73.1	78.7	84.9	86.5	85.8	84.9	80.5	74.0	66.2
	2%	WB	55.8	59.9	61.7	68.2	72.6	78.2	81.0	81.3	79.7	75.1	67.5	59.8
		MCDB	58.3	63.7	65.3	70.7	75.5	83.2	86.0	85.7	83.7	78.3	70.2	62.4
	5%	WB	52.5	56.8	59.6	65.7	70.6	76.6	80.0	80.4	78.5	72.8	65.4	56.9
		MCDB	56.0	60.1	63.3	69.2	73.4	80.8	85.3	85.4	82.3	76.0	68.6	60.7
	10%	WB	49.5	53.2	57.1	63.5	68.9	75.1	79.1	79.4	77.4	70.4	63.2	54.1
		MCDB	53.9	56.6	61.2	67.5	72.1	78.7	84.3	84.4	81.0	74.1	66.9	58.5
Mean Daily Temperature Range	5% DB	MDBR	17.0	16.3	16.1	15.1	13.2	10.4	11.4	11.1	11.3	13.2	15.2	16.8
		MCDBR	18.8	18.4	18.8	18.2	16.5	13.6	15.3	13.9	12.7	14.2	15.4	17.5
		MCWBR	13.5	13.3	12.8	11.8	9.9	7.8	6.8	6.6	7.2	8.7	10.1	12.1
	5% WB	MCDBR	15.8	15.4	15.4	13.8	12.1	11.9	11.9	11.8	10.9	10.6	12.0	14.2
		MCWBR	13.6	12.8	12.6	10.7	8.9	7.4	6.7	6.5	7.3	7.7	9.4	11.6
Clear-Sky Solar Irradiance	taub	0.409	0.454	0.515	0.538	0.550	0.554	0.539	0.526	0.508	0.473	0.424	0.397	
	taud	2.144	2.050	1.894	1.891	1.931	2.000	2.081	2.106	2.121	2.134	2.194	2.200	
	Ebn at Noon	249	250	243	244	242	240	243	245	243	241	243	246	
	Edh at Noon	40	48	60	63	61	56	52	50	47	43	37	36	
All-Sky Solar Radiation	RadAvg	863	1029	1321	1509	1555	1243	1608	1612	1320	1111	878	790	
	RadStd	68	90	88	161	179	206	193	127	151	129	91	60	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page